

Assignment 8

1. Given $L\left(\frac{1}{\sqrt{2}}(|a\rangle_A + |b\rangle_B)\right) = \frac{1}{\sqrt{2}}(|a\rangle_A + |b\rangle_A) \frac{1}{\sqrt{2}}(|a\rangle_B + |b\rangle_B)$

L.H.S : $\frac{1}{\sqrt{2}}(|a\rangle_A |0\rangle_B + |b\rangle_A |0\rangle_B) \Rightarrow |a\rangle_A |a\rangle_B + |b\rangle_A |b\rangle_B \quad - (1)$

R.H.S : (Tensor product)

$\rightarrow |a\rangle_A |a\rangle_B + |a\rangle_A |b\rangle_B + |b\rangle_A |a\rangle_B + |b\rangle_A |b\rangle_B \quad - (2)$

So, here L.H.S $\neq$ R.H.S because $|a\rangle_A |b\rangle_B$ and $|b\rangle_A |a\rangle_B$ probability is zero

Non-cloning theorem survives on the concept of not able to copying

linear combinations and linearity stands important.

2.a) $|+\rangle = \frac{1}{\sqrt{2}}(\alpha|0\rangle_A + \beta|1\rangle_A) \quad - (1)$

We are supposed to be normalizing the above it will work as general quantum teleportation algorithm.

2.b) $H_{q_2} C_{q_{21}} H_{q_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

$= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}$

2.c As it is $|+\rangle$ base

$|+\rangle_{q_1} = |+\rangle_{|0\rangle} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

create entanglement & set up the Bell-type measurement

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

after $(I \otimes H) = \frac{1}{\sqrt{2}}(|+\rangle|0\rangle + |-\rangle|1\rangle)$

measuring q_1 gives

$m = 0$: state of q_0 collapses to $|+\rangle$

$m = 1$: $|-\rangle$

Applying Z^m on q_0 flips $|-\rangle \rightarrow |+\rangle$ and leaves $|+\rangle$ unchanged

$$4. \frac{1}{\sqrt{2}} (\alpha |000\rangle + \beta |101\rangle + \alpha |011\rangle + \beta |110\rangle)$$

$$= \frac{1}{\sqrt{2}} (\alpha |0\rangle \otimes |0\rangle \otimes |0\rangle + \beta |1\rangle \otimes |0\rangle \otimes |1\rangle + \alpha |0\rangle \otimes |1\rangle \otimes |1\rangle + \beta |1\rangle \otimes |1\rangle \otimes |0\rangle)$$

$$= \frac{1}{\sqrt{2}} (\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}) + \beta \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix})$$

$$= \frac{1}{\sqrt{2}} \left(\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right)$$

$$= \frac{1}{\sqrt{2}} \left[\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha \\ 0 \\ 0 \\ \beta \\ 0 \\ \alpha \end{bmatrix}$$

6. 1) Due to Non-cloning theorem it won't allowed cloning of linear arbitrary states so we need to measure Alice.

2) Entanglement swapping

3) Default circuit represents Alice to teleport $|0\rangle$ to Bob.

So yes quantum teleportation happens by default.